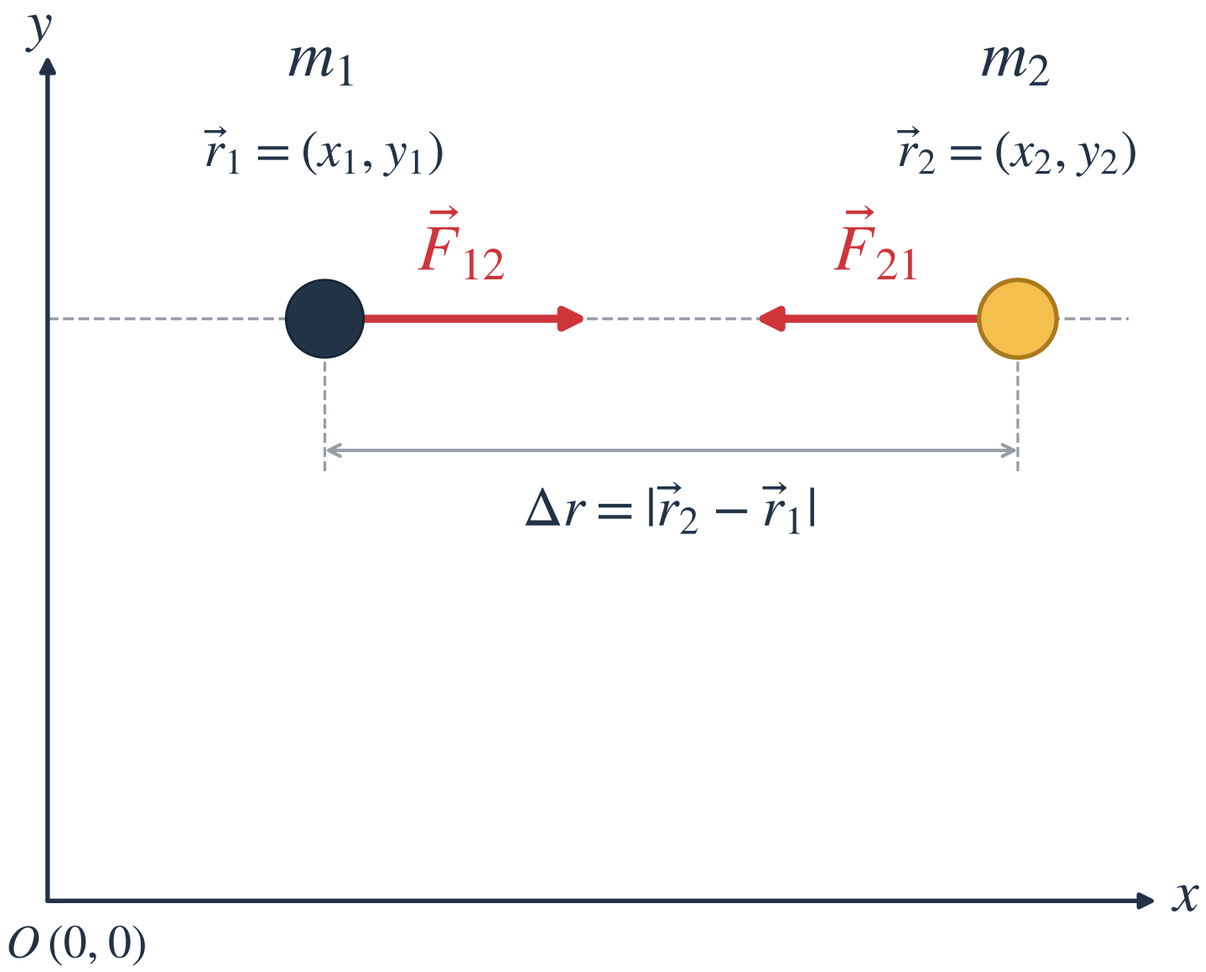
# Gravitational forces between two bodies




$$\vec{F}_{12} = -\frac{Gm_1m_2}{(\Delta r)^3}(\vec{r}_1 - \vec{r}_2)$$

Force on body 1 from body 2.

$$\vec{F}_{21} = -\frac{Gm_1m_2}{(\Delta r)^3}(\vec{r}_2 - \vec{r}_1)$$

Force on body 2 from body 1.

$$\vec{F}_{12} = -\vec{F}_{21}$$

Equal magnitude, opposite directions.

Each force acts on a different body.